

Glaucoma cell-context V2

Executed Python pilot | 6 October 2026

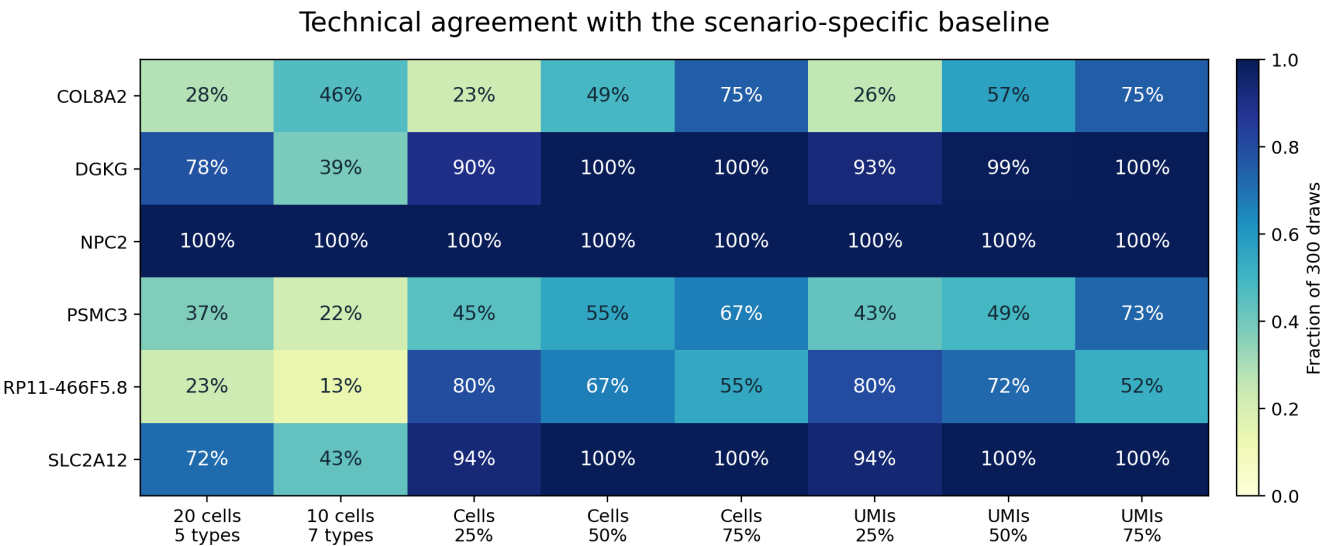
228 published gene nominations; sampling robustness; cross-study retinal RNA; predefined metabolic programs; regulatory triangulation.

Evidence layer	Executed result
RNA datasets	19,865 profiles / 3 donors; 51,645 profiles / 4 donors
Mapping	163 V1 genes; 156 external genes; 65 V1 symbols missing
RNA context agreement	78/139 in five fixed types; 70/140 in seven
Technical sampling	2,400 draws across 8 frozen scenarios
Descriptive context flag	16 genes; not a causal probability
Matched program agreement	3/6 programs share the same relative top type
Regulatory layers	25 index ATAC overlaps; 14 source gene-linked candidates

NPC2 has reproducible RNA context and a microglia-accessible index variant, with unresolved competing gene links. DGKG does not reproduce its top context; sparse genes remain weak even when a donor-omission result appears stable.

AI-assisted computational reanalysis of published healthy-donor resources. No new disease effect, causal target, fine-mapping, colocalization or MR estimate is claimed.

Sampling stability



Cell resampling within donor/type; fixed types in every draw. Technical uncertainty, not population confidence intervals.

Each draw retains the same type set as its own baseline. Primary sampling uses five eligible classes; a separate equal-10-cell sensitivity uses seven. Counts are drawn within donor/type, and all donors have equal weight. Depth thinning conserves selected-gene plus other UMIs. Percentages are technical agreement frequencies, not population confidence intervals.

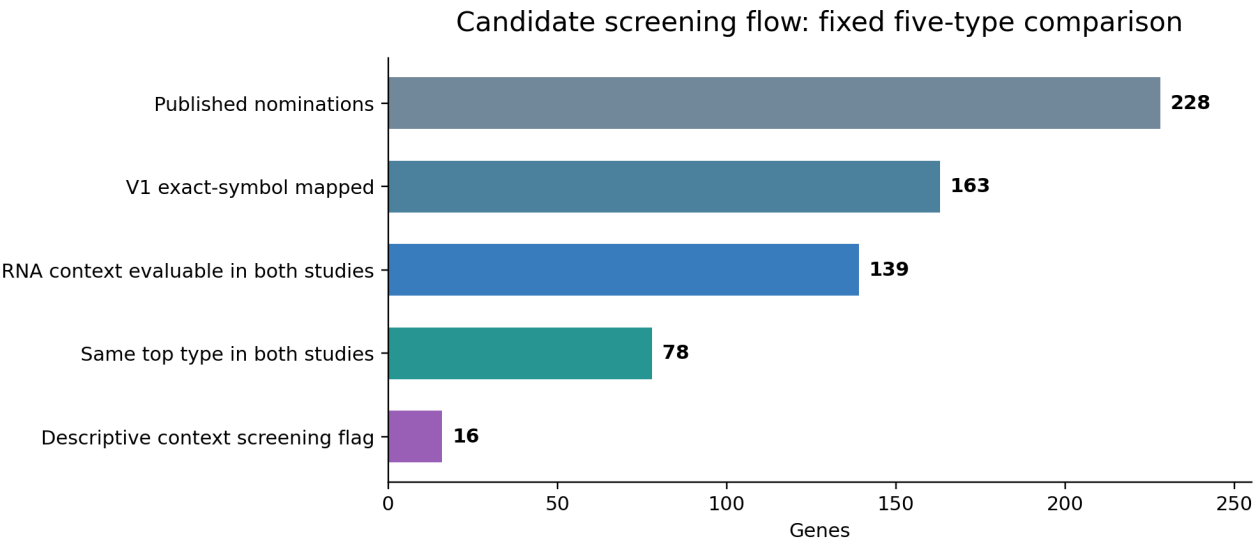
NPC2 agrees in every draw of these scenarios. COL8A2 has only six assigned-cell UMIs; many equal-20-cell draws contain no detected UMIs. PSMC3 remains unstable under equal-cell sampling despite cross-study top-type agreement.

Cross-study RNA context

Cross-study retinal RNA contexts: six prespecified cases

Candidate	V1: 5 types	Wang: 5 types	V1: 7 types	Wang: 7 types	V1 UMIs
COL8A2	RGC	Muller glia	Microglia	Microglia	6
DGKG	Muller glia	RGC	Muller glia	RGC	31
NPC2	Muller glia	Muller glia	Microglia	Microglia	3642
PSMC3	RGC	RGC	Horizontal	Horizontal	3662
RP11-466F5.8	Rod	Not mapped	Rod	Not mapped	103
SLC2A12	Muller glia	Muller glia	Muller glia	Muller glia	19

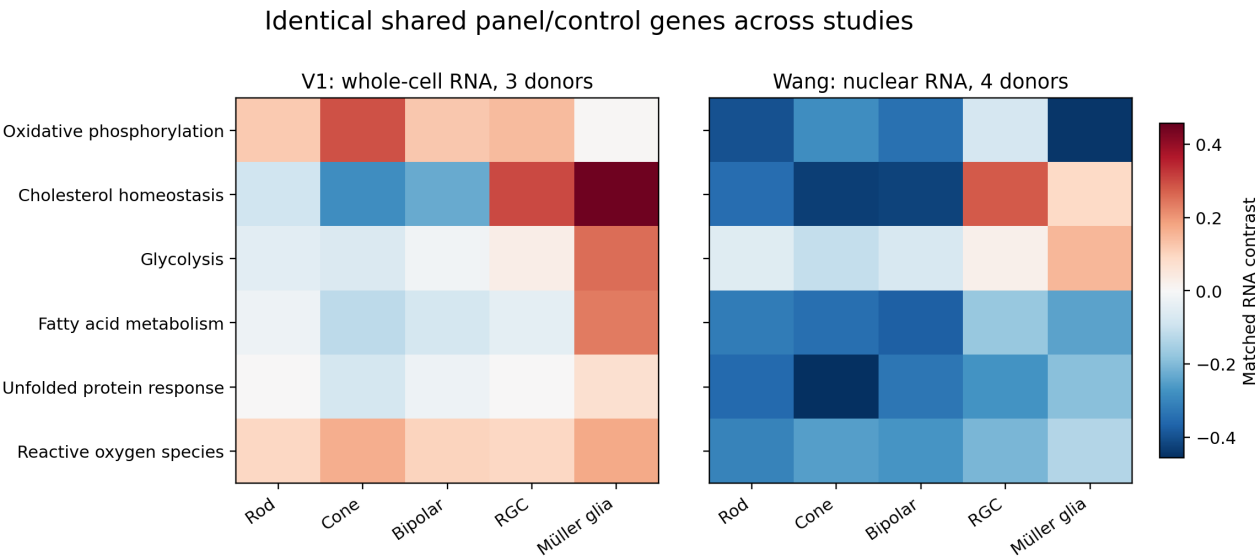
V1: 19,865 whole-cell profiles / 3 donors. Wang: 51,645 nuclear profiles / 4 donors. UMIs are V1 assigned-cell gene counts.



The screening flag is a descriptive rule; it is not statistical significance, a causal posterior, or a treatment recommendation.

Wang nuclei retain published cell labels and four donor identities. All 51,645 GEO barcodes exactly matched the curated CellxGene metadata; five validation genes matched their UMI counts cell by cell across both distributions. Eight eyes are nested within four donors. These are separate whole-cell and nuclear RNA studies, not a pooled disease comparison.

Predefined metabolic RNA programs

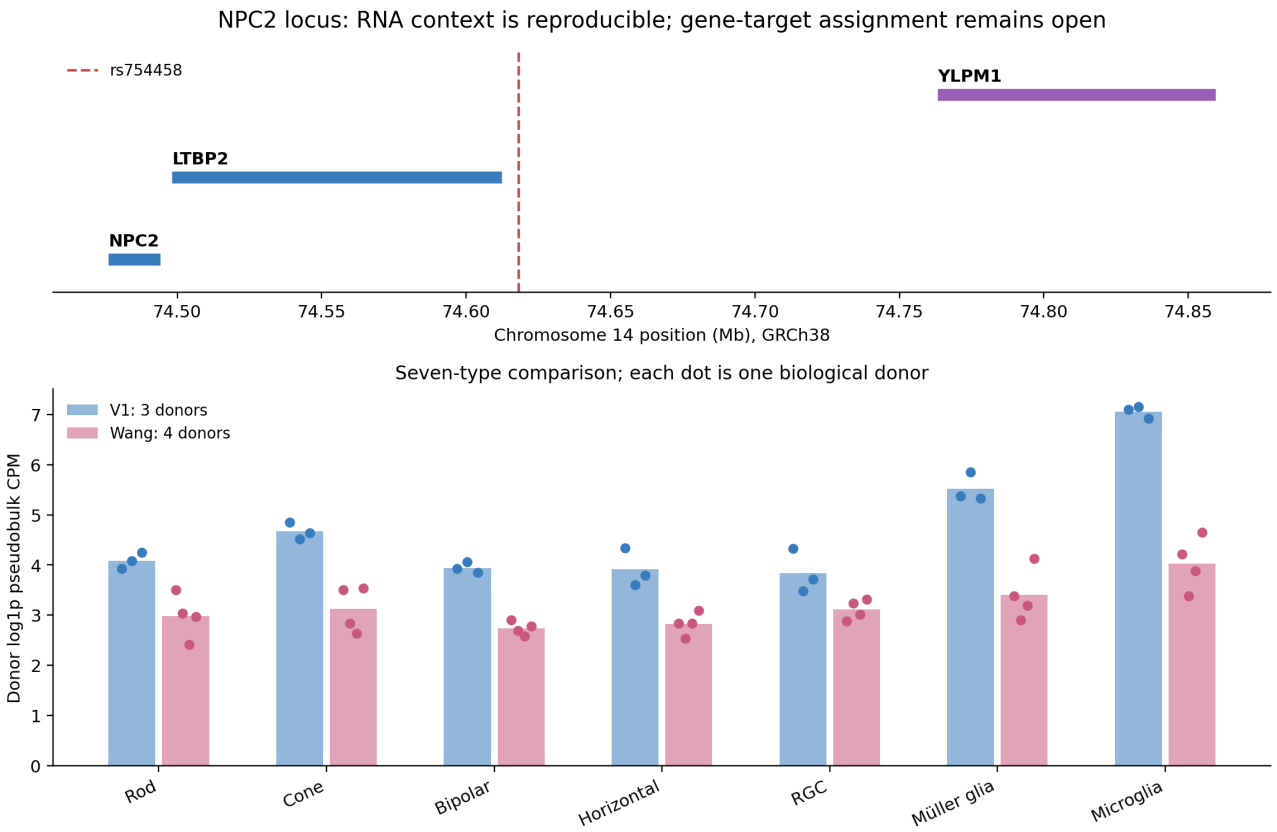


Equal-donor pseudobulk log-CPM contrasts. Relative transcriptional programs; no metabolic flux or disease effect estimates.

Both studies use the same intersected panel/control pairs and gene weights, with donor means computed separately. Only three of six programs reproduce their relative top type among five classes. The scores are transcriptional contrasts and do not measure metabolic flux. Capture chemistry, nucleus versus whole-cell sampling, donor characteristics and laboratory effects remain potential explanations for differences.

No pathway enrichment p-values, population tests, or aging/disease models were calculated from these small healthy-donor samples.

NPC2: context and competing targets



rs754458: microglia ATAC overlap; published retinal eQTL targets NPC2/LTBP2; HiChIP target YLPM1; BPNet high-effect call = 0. Retinal eQTL resources overlap across publications; these are not independent genetic replications or proven causal targets.

The rs754458 index variant intersects a microglia ATAC peak. Published retinal eQTL annotations include NPC2 and LTBP2; the published HiChIP annotation points to YLPM1. BPNet reports no high-effect call (source FDR approximately 0.882). RNA and regulatory context motivate a hypothesis; they do not resolve the causal gene or establish a microglia-specific eQTL.

Scientific scope and remaining inputs

Nearest-gene marker-peak annotations are weaker than explicit gene links. The evidence matrix keeps predicted target, promoter co-accessibility, HiChIP and retinal eQTL categories separate.

Hamel and Wang reuse overlapping published retinal eQTL resources. The independent-study RNA comparison does not make the shared genetic layer an independent replication.

Four index variants intersect source-listed cCREs from Yuan 2026, with boundary-robust overlaps. That supplied retina/macula table excludes microglia because of low cell numbers; absence there is unassessed microglia coverage.

Separate Yuan peak and gene matrices were not joined by row order without explicit pairing identifiers. That optional ABC extension needs a validated enhancer-gene edge list.

New colocalization or MR requires complete GWAS/QTL summary statistics, verified allele/build handling, LD references where needed, and adequate instrument strength. None can be reconstructed faithfully from thresholded nomination tables.

The pilot is suitable as reproducible computational evidence for an application. Publication novelty needs further benchmarking, complete genetic analyses where available, and biological validation; Q1 acceptance and citation counts are not predictable from this pilot.

Reproducibility

Each stage validates source, code, software and output hashes. Model-independent plotting uses saved estimates. V1 checkpoints are preserved; failed and temporary files are excluded from the handover. The supplied code is AI-assisted and the executed results remain subject to scientific review.

Sources

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